

FORCASTING SOC: Latent World Model Architecture :

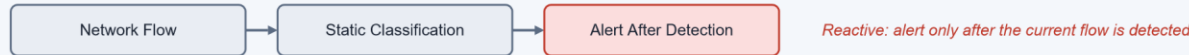
SIH26153 | **Temporal Sentinels** | Offline Predictive Cyber Defence

SYSTEM OVERVIEW & END-TO-END ARCHITECTURE

Problem Statement: SIH26153 – NTRO Domain: AI-Based Network Attack Forecasting / Predictive Cyber Defence

1 · PARADIGM SHIFT Predicts FUTURE network states instead of classifying the current flow as benign or malicious

TRADITIONAL IDS



PREDICTIVE SOC



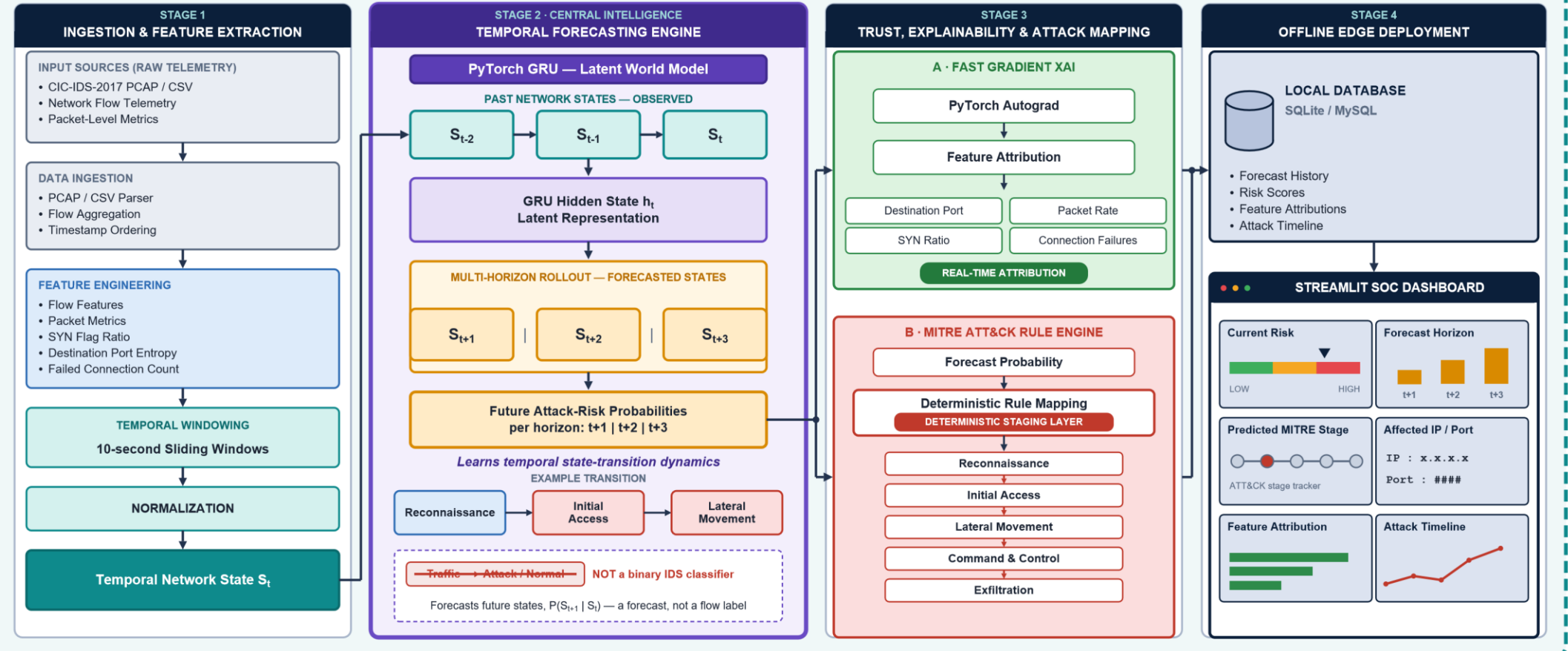
CORE LEARNING OBJECTIVE

$$P(S_{t+1} | S_t)$$

Multi-Horizon Forecast: $t+1 \rightarrow t+2 \rightarrow t+3$

AIR-GAPPED / OFFLINE DEPLOYMENT

SECURITY BOUNDARY — all components execute locally; no data leaves the perimeter



OUTSIDE THE BOUNDARY — BLOCKED

- NO CLOUD APIs
- NO EXTERNAL DATA TRANSMISSION
- NO VENDOR DEPENDENCY

DATA-TYPE COLOUR CODE

- Raw telemetry
- Processed features
- Temporal state
- Latent representation
- Forecast
- Explainability
- MITRE mapping
- Dashboard output

INFRASTRUCTURE · DEPLOYMENT PROFILE

Compute: CPU / Edge SOC Hardware

- PyTorch GRU inference
- Standard laptop / edge SOC hardware
- CPU-compatible inference
- No large-scale GPU cluster required
- Local database (SQLite / MySQL)
- Offline Streamlit deployment
- No external API dependency



Component-Level Architecture & Data Flow :

Predictive SOC: Latent World Model | SIH26153 | Temporal Sentinels

COMPONENT DEEP-DIVE :

Telemetry > Forecast > Explain > Map > Display Offline / Air-Gapped · PyTorch GRU

